Current makes a magnetic field

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect current direction with the magnetic field around a wire.

SUCCESS CRITERIA - I can...

1. Sketch circular field lines around a straight wire.
2. Use a stated right-hand rule.
3. Predict the effect of reversing current.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A total current of 12 A splits between two identical parallel branches. Find each branch current.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $B = \mu_0 I/(2\pi r)$; $\mu_0 = 4\pi \times 10^{-7}$ T·m/A.
Long straight wire. Right thumb follows conventional current; curled fingers show field direction.

Worked reasoning: Around a straight wire carrying conventional current out of the page, the magnetic field circles counterclockwise by the right-hand rule. A dot means out of page.

Your turn: Draw three field arrows around the wire and explain what the dot represents.

Marzano check - circle: ① ② ③

03 Look and explain

Conventional current points out of the page. Use the right-hand rule to draw the surrounding magnetic field direction.

Current out of page (dot). Add field arrows.



Marzano check - circle: ① ② ③

04 Build the solution

At a fixed distance, current doubles. Use the supplied relationship B proportional to I to compare field strength.

Marzano check - circle: ① ② ③

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05 Use the idea

What happens to field direction if the current reverses while its magnitude stays the same?

Marzano check - circle: 1 2 3

06 Choose a method

Two objects have charges -4 nC and -6 nC. Predict attraction or repulsion and draw both force directions.

Marzano check - circle: 1 2 3

07 Exit check

At the same current, distance from a long wire doubles. Using B proportional to 1/r, compare field strength.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Return to the worked example in problem 2. Reverse the current. Redraw the field arrows and explain what stays the same in an ideal symmetric setup.

Marzano check - circle: 1 2 3